1. Electric Current, Units, AC & DC

Definition: Electric current is defined as the rate of flow of electric charge through any cross-section of a conductor. Direct Current (DC) flows only in one direction, while Alternating Current (AC) changes its magnitude continuously and reverses its direction periodically.

In conductors like metals, current is constituted by the flow of free electrons. By convention, the direction of electric current is taken as opposite to the direction of flow of electrons.

Current (I) = Charge (Q) / Time (t)
SI Unit: Ampere (A) = Coulomb / Second

1 Ampere Definition: If 1 Coulomb of charge flows through a cross-section of a conductor in 1 second, the current is said to be 1 Ampere.

Direct Current (DC) vs Alternating Current (AC):

- **Direct Current (DC):** The current whose magnitude remains constant and direction does not change with time. *Sources:* Dry cells, batteries, DC generators. Used in electronic gadgets, electroplating.
- **Alternating Current (AC):** The current whose magnitude changes continuously with time and direction reverses periodically. *Sources:* AC generators, household power supply (220V, 50Hz in India). Used for long-distance power transmission as it causes less energy loss.

```
Direct Current (DC) Graph:          Alternating Current (AC) Graph:
   I |                               I |   Peak
   |   _________________               |    /\      /\
   |  | constant flow                  |   /  \    /  \
   |  |                               __|__/____\__/____\__ Time (t)
   |__|_________________ Time (t)      | /      \/
   0                                   |/    Trough
```

Hinglish Explanation: Current ka simple matlab hai electrons ka behna (flow hona). Ek second me jitna charge bahega, wahi current hai (I=Q/t). DC battery se milta hai jo ek hi direction me chalta hai (jaise mobile battery). AC wahi bijli hai jo hamare gharo me aati hai, jo lagatar apni direction aur value badalti rehti hai.

2. Resistance, Specific Resistance & Conductance

Definition: Resistance (R) is the property of a conductor by virtue of which it opposes the flow of electric current through it. It occurs due to the collision of free electrons with the atoms/ions of the conductor.

Factors Affecting Resistance:

The resistance of a wire depends on the following factors:

- **Length (L):** Resistance is directly proportional to the length of the conductor ($R \propto L$). A longer wire has more resistance.
- **Area of Cross-section (A):** Resistance is inversely proportional to the area of cross-section ($R \propto 1/A$). A thicker wire has less resistance.
- **Nature of Material:** Different materials have different numbers of free electrons (e.g., Copper has less resistance than Iron).
- **Temperature:** For pure metals, resistance increases with an increase in temperature.

Specific Resistance or Resistivity (ρ):

Combining the above factors, $R \propto L/A \Rightarrow$ **$R = \rho(L/A)$**. Here ρ (rho) is a constant called specific resistance or resistivity of the material.

Definition: Specific resistance is the resistance of a conductor of unit length and unit area of cross-section. It depends ONLY on the material and temperature, NOT on shape or size.

$\rho = (R \times A) / L$

SI Unit: Ohm-meter (Ω·m)

Conductance (G) & Specific Conductance (σ):

- **Conductance (G):** The reciprocal of resistance is called conductance. It measures how easily current can flow.
  $G = 1/R$. Unit: mho (℧) or Siemens (S) or Ohm^{-1}.
- **Specific Conductance / Conductivity (σ):** The reciprocal of resistivity is called specific conductance.
  $\sigma = 1/\rho$. Unit: mho/meter or Siemens/meter (S/m).

👉 **Hinglish Explanation:** Resistance (R) ka kaam current ko rokna hai (jaise raste me speed breaker). Lamba taar (wire) loge to resistance badhega, mota taar loge to resistance kam hoga. Specific resistance (ρ) taar ki lambaayi ya motayi par depend nahi karta, wo sirf ispar depend karta hai ki taar kis cheez (copper, aluminium) ka bana hai. Conductance bilkul resistance ka ulta hota hai.

3. Series & Parallel Combination of Resistances

Definition: Resistors can be connected together to achieve a desired equivalent resistance. In series, they provide a single path for current, while in parallel, they provide multiple paths.

1. Series Combination:

When resistors are connected end-to-end sequentially, they are in series. This combination is used when we need to **increase** the

total resistance of the circuit.

- The **current (I) remains the same** through all resistors.
- The total potential difference (Voltage) is the sum of voltages across individual resistors: $V = V_1 + V_2 + V_3$.
- **Formula: $R_{eq} = R_1 + R_2 + R_3 + \ldots$**
- *Disadvantage:* If one resistor fails (e.g., a bulb breaks), the entire circuit breaks.

2. Parallel Combination:

When one end of all resistors is connected to a common point and the other end to another common point, they are in parallel. This is used to **decrease** the total resistance of the circuit.

- The **potential difference (V) remains the same** across all resistors.
- The total current is the sum of currents through individual branches: $I = I_1 + I_2 + I_3$.
- **Formula: $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3 + \ldots$**
- *Advantage:* Household wiring is done in parallel so that if one appliance is turned off, others keep working independently.

```
SERIES CONNECTION:                    PARALLEL CONNECTION:
      I --->                                 I --->
  ---[ R₁ ]---[ R₂ ]---[ R₃ ]---        ----|---[ R₁ ]---|----
      V₁        V₂        V₃                |---[ R₂ ]---|
                                              |---[ R₃ ]---|
  Current is SAME. Voltage DIVIDES.     Voltage is SAME. Current DIVIDES.
```

👉 **Hinglish Explanation:** Series me resistors ek ke baad ek lage hote hain. Isme Current (I) same rehta hai, aur equivalent resistance plus hota hai (R=R1+R2). Parallel me resistors upar-neeche lage hote hain, isme Voltage (V) same rehta hai, aur equivalent resistance (1/R) ke form me judta hai jisse total resistance kam ho jata hai. Ghar ki wiring parallel me hi hoti hai.

4. Carbon Resistances and Colour Coding

Definition: Carbon resistors are compact, inexpensive, and widely used fixed resistors made from carbon composition. Because they are too small to print resistance values on them, a standard colour-coding system is used to indicate their value and tolerance.

A typical carbon resistor has 4 coloured bands printed on its body.

- **First Band:** Indicates the first significant digit.
- **Second Band:** Indicates the second significant digit.
- **Third Band (Multiplier):** Indicates the power of 10 by which the first two digits are multiplied.
- **Fourth Band (Tolerance):** Indicates the possible percentage variation (error) in the resistance value. Gold = ±5%, Silver = ±10%, No Color = ±20%.

Colour Code Mnemonic (Very Important):

B B R O Y of **G**reat **B**ritain had a **V**ery **G**ood **W**ife.

Colour	Digit	Multiplier

Black	0	10^0
Brown	1	10^1
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Grey	8	10^8
White	9	10^9

Example: A resistor has bands: Red, Violet, Orange, Gold.
Red = 2, Violet = 7, Orange = 10^3, Gold = ±5%.
Value = **27×10^3 Ω ± 5%** or 27 kΩ ± 5%.

👉 **Hinglish Explanation:** Carbon resistors chote hote hain aur TV/Radio ke circuit board me use hote hain. Inke upar value likhne ki jagah colors ki 4 pattiyan (bands) bani hoti hain. Shuru ke do color numbers batate hain, teesra zeroes batata hai (multiplier), aur chautha tolerance (accuracy). "BB ROY..." wali trick se numbers 0 se 9 tak yaad rakhne hain.

5. Ohm's Law and Its Verification

Definition: Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference applied across its ends, provided the physical conditions (like temperature, strain, etc.) remain constant.

Mathematical Expression:

$$V \propto I$$
$$V = I \times R$$

Where 'V' is Potential Difference (Volts), 'I' is Current (Amperes), and 'R' is the proportionality constant called Resistance (Ohms).

Verification of Ohm's Law:

To verify Ohm's law practically in a lab, we need a battery, a voltmeter, an ammeter, a rheostat (variable resistor), a switch, and an unknown resistor (wire).

- The **Ammeter** is connected in **series** with the resistor to measure current (I).

- The **Voltmeter** is connected in **parallel** across the resistor to measure voltage (V).
- The Rheostat alters the current in the circuit.

Procedure: The circuit is closed, and for different settings of the rheostat, readings of V and I are recorded. When a graph is plotted between V (y-axis) and I (x-axis), it forms a **straight line passing through the origin**. The slope of this V-I graph gives the resistance (R) of the conductor. This straight line verifies that $V \propto I$.
Limitations: Ohm's law is not universal. Devices like diodes, transistors, and electrolytes do not obey Ohm's law (Non-ohmic conductors).

```
Circuit Diagram for Verification:
        +-----(A)-----[ Resistor R ]-----+
        |                 |      |       |
      [Battery]           +-(V)--+     [Rheostat]
        |                                |
        +-----------[ Switch ]-----------+

V-I Graph:
    V |      /
      |    /   Slope = Resistance (R)
      |  /
      |/_______________ I
```

👉 **Hinglish Explanation:** Ohm's law sabse basic law hai jo kehta hai: jitna zyada Voltage (battery ki takat), utna zyada Current (V = IR). Ye tabhi kaam karta hai jab taar ka temperature change na ho. Lab me isko verify karne ke liye Voltmeter aur Ammeter ka use karte hain. V aur I ka graph banate hain, agar bilkul seedhi line aati hai to Ohm's law verified!

6. Kirchhoff's Laws

Definition: Kirchhoff's laws are two fundamental rules used to analyze complex electrical circuits where Ohm's law alone is not sufficient to find current and voltage in different branches.

1. Kirchhoff's Current Law (KCL) / Junction Rule:

Statement: The algebraic sum of currents meeting at a junction (node) in an electrical circuit is always zero. ($\Sigma I = 0$).
In simpler terms: Total current entering a junction = Total current leaving the junction.

- **Sign Convention:** Current entering is taken as positive (+), and current leaving is taken as negative (-).
- **Principle:** It is based on the **Law of Conservation of Charge** (charge cannot accumulate at a point).

2. Kirchhoff's Voltage Law (KVL) / Loop Rule:

Statement: In any closed loop (mesh) of an electrical circuit, the algebraic sum of all the electromotive forces (EMFs) and the voltage drops (IR) is equal to zero. ($\Sigma E + \Sigma IR = 0$).

- **Sign Convention:**
  - Traversing in the direction of current: IR drop is negative.
  - Traversing from positive to negative terminal of battery: EMF is negative (and vice-versa).
- **Principle:** It is based on the **Law of Conservation of Energy**.

👉 **Hinglish Explanation:** Complex circuits solve karne ke liye 2 laws hain. First law (KCL) kehta hai ki kisi bhi point (junction) par jitna current aayega, utna hi current waha se jayega (Charge conservation). Second law (KVL) kehta hai ki ek band raste (closed loop) me saari batteries ka voltage aur resistors ka voltage drop milakar total zero hota hai (Energy conservation).

7. Wheatstone Bridge & Slide Wire Bridge

Definition: Wheatstone bridge is an electrical circuit arrangement consisting of four resistors used to accurately measure an unknown resistance. The Slide Wire Bridge (Meter Bridge) is its practical application.

Wheatstone Bridge Principle:

It consists of four resistors P, Q, R, and S connected to form a quadrilateral. A galvanometer is connected between two opposite junctions, and a battery between the other two.
Balanced Condition: When the bridge is balanced, no current flows through the galvanometer (deflection is zero). In this state, the ratio of resistances in adjoining arms is equal.

$$P / Q = R / S$$

If P, Q, and R are known, the unknown resistance S can be calculated as S = R × (Q/P).

Slide Wire Bridge (Meter Bridge):

It is the practical form of the Wheatstone bridge used in labs. It consists of a 1-meter (100 cm) long uniform resistance wire (usually manganin or constantan) stretched on a wooden board. A jockey slides over this wire to find the null point (balancing length 'l' where galvanometer shows zero).

- Let the balancing length from point A be **l cm**.
- The remaining length is **(100 - l) cm**.
- Resistance of length 'l' acts as P, and length (100-l) acts as Q. Known resistance box is R, and unknown wire is S.

Applying the balanced condition (P/Q = R/S):
l / (100 - l) = R / S

$$\text{Unknown Resistance (S)} = R \times [(100 - l) / l]$$

👉 **Hinglish Explanation:** Wheatstone bridge 4 resistors ka ek network hai. Jab ye "balance" hota hai to beech wale raste (galvanometer) se current nahi jata, aur formula (P/Q = R/S) lagta hai. Slide Wire Bridge (Meter bridge) isi principle par kaam karne wala lab apparatus hai jisme 1 meter lamba taar hota hai. Taar par jockey ghuma kar null point (zero current) dhundhte hain aur anjaan taar ka resistance nikalte hain.

8. EMF and Terminal Potential Difference

Definition: EMF is the maximum potential difference between the terminals of a cell when no current is being drawn from it (open circuit). Terminal Potential Difference is the voltage across the terminals when current is flowing (closed circuit).

Every cell/battery has an **Internal Resistance (r)** offered by its own electrolyte and electrodes. This causes a voltage drop inside the cell when it is used.

Electromotive Force (EMF - denoted by E)	Terminal Potential Difference (TPD - denoted by V)
It is the potential difference across terminals when the cell is in an **open circuit** (I = 0).	It is the potential difference across terminals when the cell is in a **closed circuit** (I > 0).
It is a characteristic of the cell and remains constant.	It depends on the amount of current drawn from the cell.
EMF is always greater than TPD (while discharging).	TPD is always less than EMF (V = E - Ir).

$$V = E - I \cdot r$$

Where E = EMF, V = Terminal Voltage, I = Current drawn, r = Internal resistance.
Note: When a cell is being charged, the current flows in the reverse direction, making V = E + Ir.

👉 **Hinglish Explanation:** EMF (Electro Motive Force) battery ki asali maximum power hai jab battery sirf rakhi hui hai aur usme se current use nahi ho raha. Jaise hi aap switch on karke current lete ho, battery ke andar hi thodi takat waste ho jati hai (internal resistance ke karan). Ab jo thodi kam voltage bahar milti hai, use Terminal Potential Difference (V) kehte hain. (V hamesha E se thoda kam hota hai).

9. Heating Effect, Power, Energy & Advantages

Definition: When an electric current passes through a high resistance wire (like nichrome), the electrical energy is converted into heat energy. This is called the heating effect of electric current (Joule's Heating).

Joule's Law of Heating:

The heat (H) produced in a conductor is directly proportional to the square of current (I^2), resistance (R), and the time (t) for which current flows.

$$\text{Heat (H)} = I^2 \cdot R \cdot t \text{ (in Joules)}$$

Applications: Electric iron, heater, toaster, electric fuse.

Electric Power (P):

It is the rate at which electrical energy is consumed or dissipated in a circuit.
Formulas: $P = V \cdot I = I^2 \cdot R = V^2 / R$

SI Unit: Watt (W). 1 kW = 1000 W. 1 Horsepower (HP) = 746 W.

Electric Energy (E):

It is the total work done by the current in a given time.

Energy (E) = Power × Time = P × t

Commercial Unit: Kilowatt-hour (kWh), commonly known as a 'Unit' in electricity bills.

1 kWh = 1 kW × 1 hour = 1000 W × 3600 s = 3.6×10^6 Joules.

Numerical Tip: Cost of electricity = (Power in kW) × (Time in hours) × (Rate per unit).

Advantages of Electrical Energy over Other Forms:

- **Easy Transmission:** Can be transmitted over long distances using wires with minimal power loss (using AC and transformers).
- **High Efficiency & Cleanliness:** It produces no smoke, poisonous gases, or ash at the point of consumption.
- **Easy Conversion:** Can be easily converted into light (bulbs), heat (heaters), or mechanical energy (motors).
- **Controllability:** Can be started, stopped, and regulated instantly and safely using simple switches and regulators.

> 👉 **Hinglish Explanation:** Jab current kisi taar se guzarta hai to collision ki wajah se wo garam ho jata hai, ise Heating effect kehte hain (Formula: $H=I^2Rt$, Heater me use hota hai). Power (Watt) ka matlab hai kitni tezi se energy use ho rahi hai ($P=VI$). Jo hamare ghar ka bijli bill aata hai wo KWh ('Unit') me aata hai. 1 Unit matlab 1000 Watt ka appliance 1 ghante tak chalana. Electrical energy best hai kyuki isme dhuaan nahi hota aur control karna asaan hai.

🔁 Quick Revision Notes (Exam Saver)

- **Current & Units:** $I = Q/t$. Unit: Ampere. DC is constant, AC alternates.
- **Resistance & Resistivity:** $R = \rho(L/A)$. ρ depends only on material and temp.
- **Resistor Combos:** Series $R = R_1+R_2$ (Current same). Parallel $1/R = 1/R_1+1/R_2$ (Voltage same).
- **Color Code:** BB ROY Great Britain... Multiplying power increases by 10. Gold ±5%, Silver ±10%.
- **Kirchhoff's Laws:** KCL ($\Sigma I=0$, Junction) -> Charge conserved. KVL ($\Sigma V=0$, Loop) -> Energy conserved.
- **Meter Bridge Formula:** $S = R(100-l)/l$. Works on Wheatstone principle ($P/Q = R/S$).
- **EMF vs TPD:** EMF is open circuit voltage. TPD is closed circuit voltage ($V = E - Ir$).
- **Power & Energy:** $P = VI$. Energy = Pt. 1 kWh = 3.6×10^6 Joules (1 Unit of electricity bill).

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